

YUEMENG LI

CONTROL ENGINEERING / AUTOMATION / AI TOOLS

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Master's student at Beijing University of Chemical Technology with a background in automation. Hands-on experience in microchip research, production-line controls and project support. Built projects in computer vision, traffic-light control and AI-assisted investment research.

EDUCATION

Beijing University of Chemical Technology

Sep 2024 - Jul 2027
(expected)

Master's in Control Science and Engineering | Admitted by recommendation

- GPA: 3.53/4.00; top 5%. Courses include sensing technology, microelectromechanical systems (MEMS), and system modeling.

Beijing University of Chemical Technology

Sep 2020 - Jun 2024

Bachelor's in Automation

- GPA: 3.52/4.33; top 5%. Courses include automatic control, modern control theory, process control, and analog and digital electronics.

INTERNSHIPS

AAC Technologies

Jun 2026 - Jul 2026

Project Management Intern

- Supported a new microphone module project. Updated schedules and tracked design, materials, manufacturing, equipment and production-line readiness.
- Organized trial-production and quality-test data, and followed up on risks and open issues. Supported the project as the sample scale grew from about 11,800 to 20,000.

Foxconn Technology Group

Jul 2022 - Aug 2023

Anodizing Line Control Intern

- Used chemical concentration data to develop automatic dosing logic, reducing concentration fluctuations by about 16% and lowering the need for manual dosing.
- Helped test the automated crane that moves products between chemical baths, checking its routes and stop times.

Shenzhen Yuansen Optoelectronics Technology Co., Ltd.

Jul 2021 - Aug 2021

Product Intern

- Compared features, performance and prices for over 30 LED and optical products in Europe. Studied how price changes could affect demand.
- Suggested product improvements and helped develop pricing and promotion plans for the European market.

SKILLS

Programming and AI: Python, MATLAB, C, Linux, pi-ai, AI agent workflows.

Engineering: PLC programming, PID tuning, MEMS fabrication, L-Edit, AutoCAD.

Data and tools: Cutadapt, STAR, featureCounts, MA/MACD/RSI, Office, WPS.

Languages and certificates: English (CET-4 and CET-6); National Computer Rank Examination, Level 2.

PROJECTS & ACHIEVEMENTS

SELECTED PROJECTS

AI Investment Research and Decision-Support Assistant

2025 - Present

Independent Developer | Python, pi-ai, language models

- Designed a personal research system with Research, Decision and Report agents to automate market research, decision support and report writing.
- Combined stock and ETF prices, financial news and policy updates. Used MA, MACD and RSI with a language model to assess market trends over different time frames and suggest portfolio changes.
- Built tools that account for current holdings, available funds and risk limits. Added position-size suggestions, tests on historical data, decision records and an interactive web interface.

Microchips for Single-Cell RNA Research

Sep 2024 - Present

Project Lead | MEMS, PDMS, Python, Linux

- Designed chip layouts, made molds, cast PDMS chips and tested their performance to collect RNA from individual cells. Improved the design and experiments based on results.
- Wrote Python scripts and used Cutadapt, STAR and featureCounts to clean sequencing reads, match sequences and measure gene activity.
- Filed a Chinese invention patent application, no. 2025115308827.

Adaptive Traffic Lights with PLC Control

Sep 2023 - Oct 2023

Project Lead | PLC, digital I/O, timers and counters

- Programmed traffic-light sequences for both road directions. Configured digital inputs and outputs for vehicle sensors, pedestrian buttons and signal lights.
- Used timers, counters and interlocks to handle each pedestrian request in a single crossing cycle, switch signals safely and count pedestrian requests.
- Adjusted green lights based on vehicle detection. Added fixed-timing and sensor-based modes to respond to current traffic conditions.

Smart Camera Tracking for Basketball Broadcasts

Nov 2022 - Dec 2023

Project Lead | Python, YOLOv7, camera-mount control

- Built a labeled basketball dataset, trained a model to detect players and the ball, and deployed it on a mobile device.
- Designed a camera-target selection method and sent target coordinates wirelessly to a camera-mount controller. Achieved over 96% detection accuracy with average detection latency below 50 ms.
- Recognized as a national-level undergraduate innovation project; won third prize in the Beijing division of the China International College Students' Innovation Competition.

AWARDS & CAMPUS LEADERSHIP

2024-2025: Special-grade and First-grade Graduate Academic Scholarships, BUCT.

2025: BUCT-Wanji Technology Scholarship.

2024: China International College Students' Innovation Competition, Beijing Third Prize.

Sep 2025 - Aug 2026: Led the Academic and Technology Department of the graduate student association.

Planned academic and technology events, coordinated resources and worked with different teams.